

LIZARDS

Typically, lizards have four legs and a long, thin tail, but there are also many legless species. They have scaly skin and firm jaw articulation.

All lizards are ectothermic, obtaining heat energy from the environment. Although seen as predominately tropical or desert animals, they have a global distribution; lizards are found from beyond the Arctic Circle in Europe to the tip of South America. Highly adaptable, they occupy a wide range of terrestrial habitats with many being arboreal or rock-dwelling. Many legless species are adapted for burrowing, a few lizards are effective gliders from trees, and others are semi-aquatic, including a marine species on the Galapagos Islands.

STRATEGIES FOR SURVIVAL

Lizards range in length from about 1.5 cm ($\frac{1}{2}$ in) to 3 m (10 ft) in the case of the Komodo dragon, but the majority are between 10 and 30 cm (4–12 in). Although most are carnivores, around two per cent of species are primarily herbivorous. Many lizards are prey for other carnivores. Their

defence mechanisms depend on agility, camouflage, and bluff. Many species are able to shed their tail to distract the attention of predators as they escape. The tail grows back. In some families of lizard, the skin can change colour. This ability can be used for camouflage or for sexual or social signalling. Unusually, in many lizards the pineal gland on the top of the head acts as a light-sensitive “third eye”.

VARIED LIFESTYLES

Although some species are solitary, many lizards have complex social structures, with the males maintaining their territories through visual signals. Many species lay eggs in underground nests, while others retain the eggs in the oviduct until hatching. There are also truly viviparous species, in which the mother provides nutrition via a placenta. Males have paired hemipenes — sexual organs used for internal fertilization — although some species are parthenogenetic, with females reproducing without the participation of a male. Some lizards tend their eggs during incubation but very few exhibit any maternal care of the hatchlings.

PHYLUM	CHORDATA
CLASS	REPTILIA
ORDER	SQUAMATA
FAMILIES (LIZARDS)	38
SPECIES	6,687

DEBATE

SQUAMATA: AN ALL-INCLUSIVE ORDER

Traditionally, lizards and snakes were considered two distinct groups, but modern genetic research makes it clear that they are not. Primitive squamates (scaled reptiles) were lizard-like animals that arose in the mid-Jurassic. Available evidence suggests that snakes evolved from lizards during the mid-Cretaceous. By this time various families of lizards had branched off on their own evolutionary course. Some lizard families are thus closer to snakes than they are to other lizards. Of living lizards, the monitors are thought to be those most closely related to snakes.

CHAMELEONS

Restricted to the Old World, the members of the family Chamaeleonidae have long limbs with grasping feet and a prehensile tail for gripping branches — an adaptation for life in trees. Their eyes are capable of moving independently in any direction to locate insects or small vertebrates. They catch their prey with a long, sticky tongue that shoots out from the mouth. Their ability to change colour is for display and camouflage. Many species are threatened with extinction.

8 cm
3 $\frac{1}{4}$ in

BEARDED PYGMY-CHAMELEON

Rieppeleon brevicaudatus

This unusual, small chameleon is from E. Africa. Drab coloration and patterning along its body allow it to resemble a dead leaf.

PARSON'S CHAMELEON

Calumma parsonii

The largest chameleon in the world, this species is restricted to Madagascar. It hunts invertebrates in the canopy of montane forests.

70 cm
28 in

30 cm
12 in

prehensile tail

MEDITERRANEAN CHAMELEON

Chamaeleo chamaeleon

This chameleon spends its time in bushes searching for insects. It is found in N. Africa and around the Mediterranean.

JACKSON'S CHAMELEON

Trioceros jacksonii

This diurnal, arboreal species is found in E. Africa. The male is identifiable by the three horns on its snout, which are for display.

VEILED CHAMELEON

Chamaeleo calyptratus

This species is from the southern coast of the Arabian Peninsula. Males have a large casque on the head, while females have a smaller one.

green skin coloration
sometimes changes to brown

30 cm
12 in

dorsal
spines

51 cm
20 in

GIANT SPINY CHAMELEON

Furcifer verrucosus

This large chameleon is a native of the humid, coastal region of Madagascar. A shy species, it relies on camouflage to ambush its insect prey.

56 cm
22 in

grasping,
four-toed feet

PANTHER CHAMELEON

Furcifer pardalis

Found only in dry forests in Madagascar, this lizard hunts insects in trees by stealth. Males are highly territorial.

60 cm
23 $\frac{1}{2}$ in